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hardware. Progress in robotics especially, depends on the development of cost-effective, innovative solutions using new-generation OSH platforms and rapid fabrication tools,” says T Jayakrishnan, director and country manager, Energid Robotics and Machine Vision.

OSH, with its characteristic features of flexibility, modularity and reusability, would help do just that. The wheel never needs to be reinvented, and you can easily use open source intellectual property (IP) as building blocks for a new project. This kind of reuse reduces cost, and also speeds up the process of converting an idea into a finished product—especially the initial phases of feasibility analysis and proof-of-concept demonstration. It also fosters research and learning with little investment, and encourages innovation.

Community support: One individual can never do as much as a community! So, it is always better to open up a system for a whole community to develop and improve. “Making hardware open-source helps foster a huge support community. If someone has already done something, users just have to plug-and-play, or contribute to the system and help it grow bigger and bigger,” says Azad.

Common standards: A natural consequence of community development is the evolution of common standards, making it easy for other systems to integrate seamlessly with an open-source technology.

Low-cost manufacturing: OSH makes not just component designs available, but also the knowledge required to convert raw components into good-quality finished products. This enables not just innovation and further improvement of product designs, but also low-cost manufacturing. People in any corner of the world, who think they have the resources and bandwidth to manufacture a product, can seek the required permissions and manufacture it based on the open-source designs and procedures. This kind of localised manufacturing would help bring down the cost of the components in various geographies.

Easy hardware-software integration: It has also been found that due to the availability of complete specifications and the modular nature of OSH, it is easier to integrate hardware and software. Since no robotics project is complete, or even possible, without complex software systems today, this is a major benefit.

A multi-disciplinary approach: A low-cost, Web-based community development model encourages many others to contribute. “When it is free, people are more willing to experiment with, and try out, something that they would not ordinarily have done. OSH allows absolutely anyone and everyone to contribute—even those who may not be engineers. If they have free access to some hardware, then even a biologist or an economist can chip in. So, OSH encourages the fusion of various streams of knowledge,” says de Sa.

Just right for learners

OSH also happens to be a superb fit for learning kits. Azad explains why this is so: “The main aim of learning is to tinker around and see what a student can do with hardware. If everything is open, depending on the interest levels of the students, they can go as deep as they want to, to change the system or develop modules for the existing system. To make this happen, the students will have to understand the entire system, and how it functions. Without open systems, every batch of students would do the same task over and over again. With open-source kits, the students can add value, which could be shared all over the world, and when the next batch comes in, they can pick it up from there. In this process, if anyone wants to go back and change something, the files and data are always available, as it is open source.”

Motivating examples

A lot of well-known robots have been, and are being built using OSH. ACyut, BITS-Pilani’s humanoid robot that went on to win the RoboGames last year, is a brilliant example. The RobotCub project (<http://www.robotcub.org/>), funded by the European Commission to build iCub, a toddler-like humanoid robot, is also based on an open framework. Vineet Sahu, another enthusiastic RoboMSR champion, points out that, “The ASIMO, which was a closed-source project, took almost 15 years to develop to the stage that it is at today, while the iCub project developed by the European Union as an open-source project took just six years to finish. This is a perfect example of the difference between closed and open source projects.”

Spykee, the network-controllable mobile robotic toy, is also advertised as an open-source robot, which can be integrated with other projects or modified to a user’s needs. E-puck, the education robot originally developed at the École Polytechnique Fédérale de Lausanne, Switzerland, for micro-engineering education, is also a classic piece of open engineering. The hardware designs as well as the on-board software of the seven-centimetre-tall differential-wheeled mobile robot are open source, and it is being built and sold by several companies now.

The Humanoid Robot project of Sweden is another example of open source, which has contributed greatly to the development of robot learning techniques.